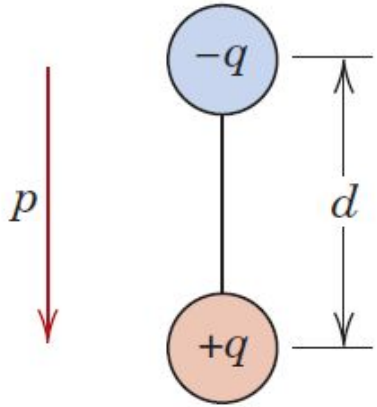
EE2021 -
Materials Science for Electrical Engineers

Dielectric Properties

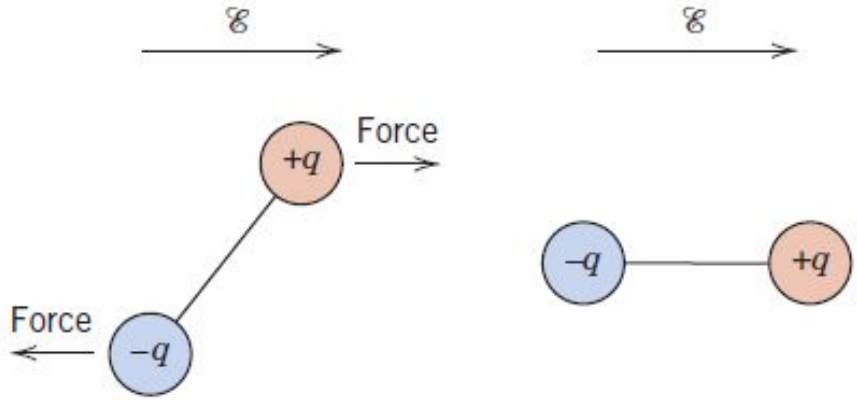
Deleep R. Nair & Sayak Dutta Gupta
Department of Electrical Engg, IIT Madras

What is a Dielectric?

- **Dielectric:** electrical insulator that can be polarized by applying electric field
- Polarization: process of electric dipole formation → separation of positive and negative charged entities at a molecular or atomic level on application of electric field
- Dielectric materials are used in capacitors



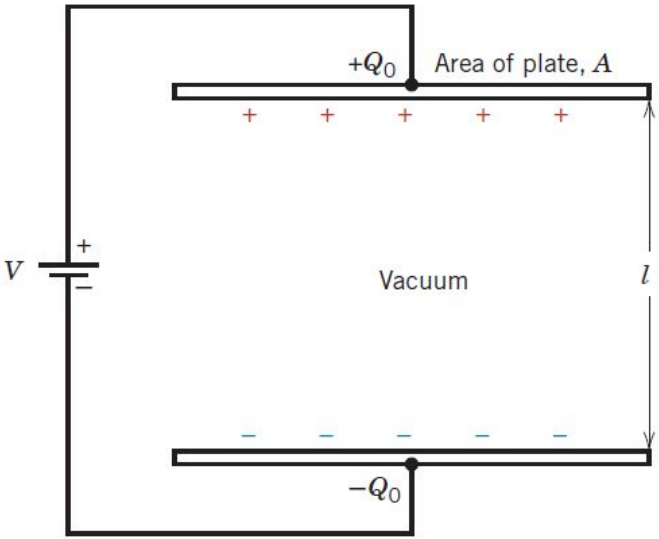
p: electric dipole moment, **q:** magnitude of each dipole charge, **d**: distance of separation



Dipole alignment with field

Capacitance

- Capacitance C is related to the quantity of charge stored on either plate Q for a voltage V applied across the capacitor by **C = Q/V**
- C is coulombs per volt, or farads (F)



For vacuum between two parallel plates

$Q_0 = D_0.A$

$= \epsilon_0 E.A$

$= \epsilon_0 (V/l).A$

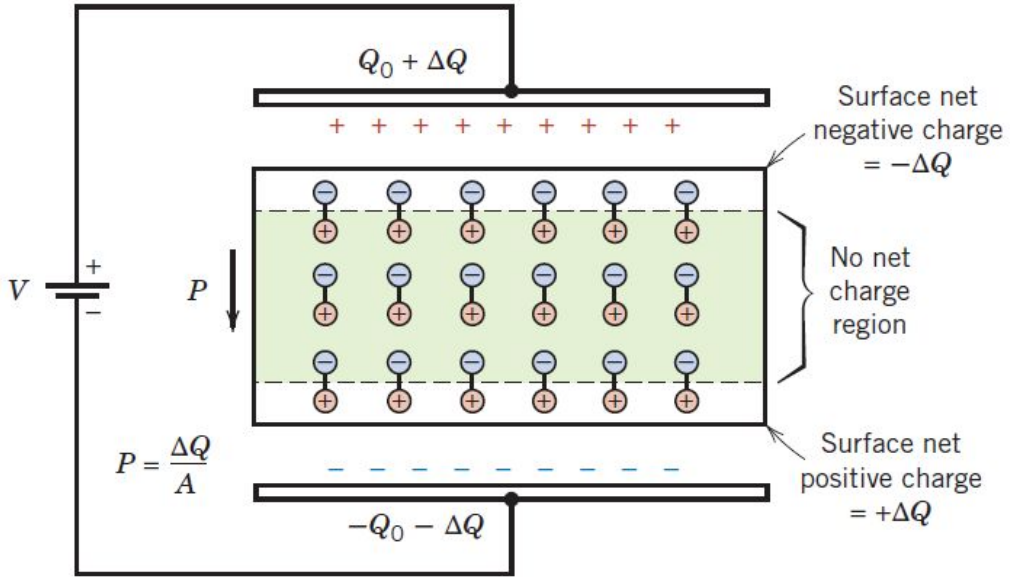
$C_0 = Q_0/V = \epsilon_0 A/l$

$+Q_0$ & $-Q_0$: charge stored on top and bottom plate

D_0: Dielectric displacement (surface charge density) in vacuum

ϵ_0 : permittivity of vacuum 8.85×10^{-12} F/m

Capacitance with dielectric



- $Q = Q_0 + \Delta Q$
  $= \epsilon_0 E.A + P.A$
  $= (\epsilon_0 E + P)A$
- $Q = D.A = \epsilon E.A = \epsilon (V/l).A$
- $C = Q/V = \epsilon A/l$
- $\epsilon_r = \epsilon/\epsilon_0$
- $D = \epsilon_0 E + P = \epsilon_r \epsilon_0 E$
- $P = \epsilon_0(\epsilon_r - 1)E$
- $\epsilon_r = (\epsilon_0 E + P)/\epsilon_0 E = 1 + \chi$

+Q & -Q: charge stored on top and bottom plate

D: Dielectric displacement,

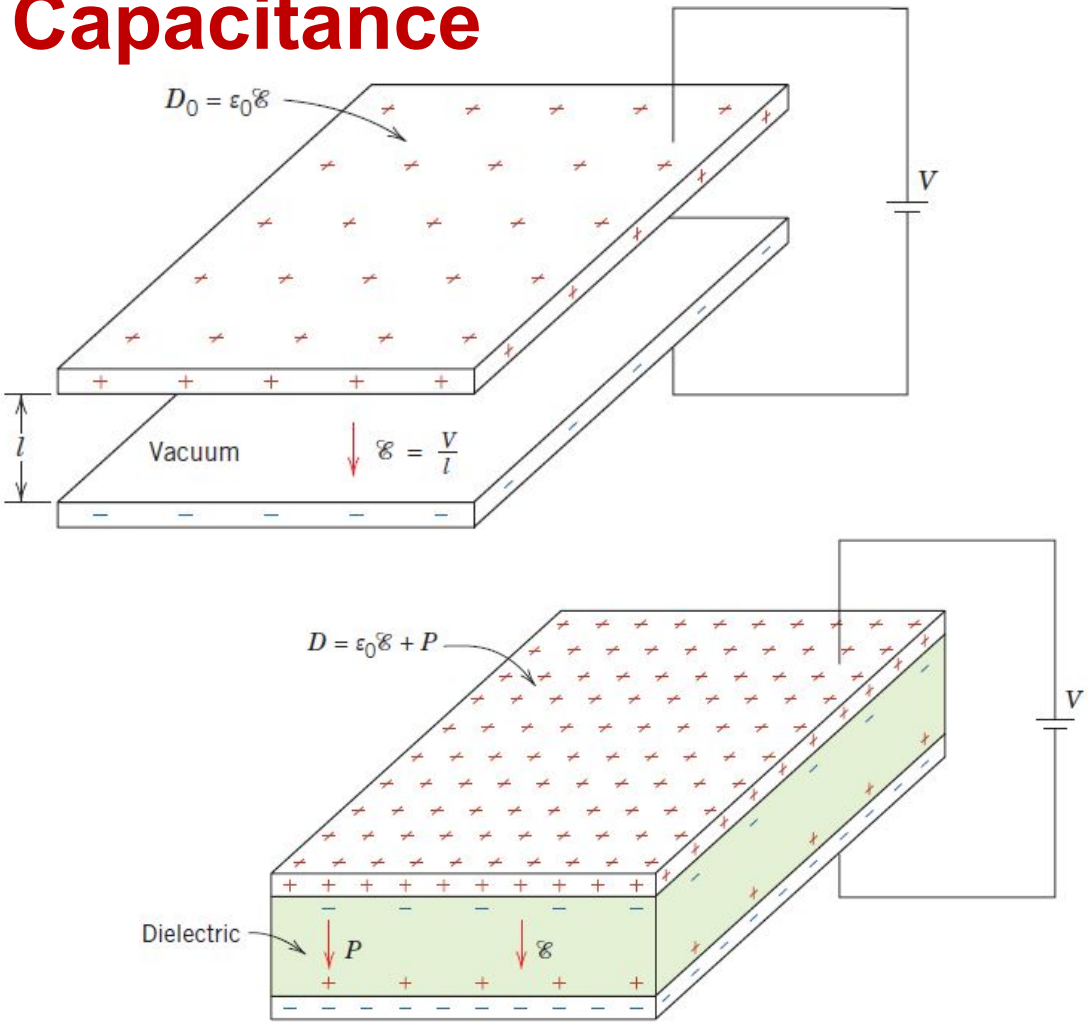
χ : Electric susceptibility (how easily a dielectric can become polarized under E)

P: Polarization (total dipole moment per unit volume) = ΔQ.l/A.l = ΔQ/A

ϵ_r : relative permittivity or dielectric constant

Non-linear and anisotropic dielectrics : P and E aren't linearly related by a scalar

Capacitance



Composition	Dielectric constant
SiO_2	3.9
Al_2O_3	9
ZrO_2	29
HfO_2	25
$HfSiO_4$	11
Ta_2O_5	26
La_2O_3	30
$LaAlO_3$	30
Nb_2O_5	35
TiO_2	95
$BaTiO_3$	1700
$SrTiO_3$	2000
$Pb(Zr,Ti)O_3$, $(Pb,La)(Zr,Ti)O_3$	2500
$CaCu_3Ti_4O_{12}$	80 000

DOI: 10.1039/C3TA14034B

Types of Polarization

Electronic polarization:

- Found in all dielectric materials
- Results from displacement of center of negatively charged electron cloud relative to positive nucleus of an atom by electric field
- Exists only while an electric field is present

Ionic polarization:

- Occurs only in materials that are ionic
- Applied field displaces cations in one direction and anions in other → net dipole moment

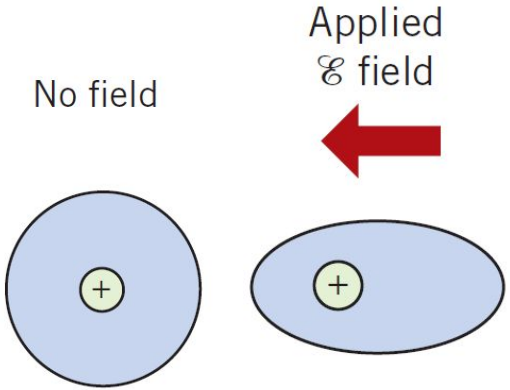
Orientation polarization:

- Found only in substances that possess permanent dipole moments
- Due to rotation of permanent moments into the direction of the applied field
- Counteracted by thermal vibrations of atoms → polarization decreases with increasing temperature

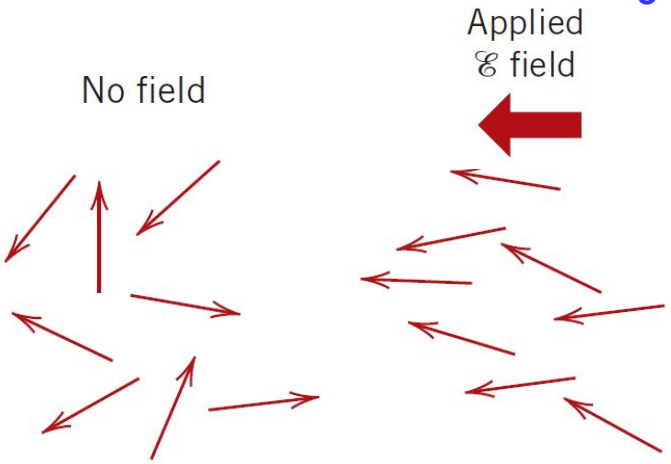
Total Polarization

$$P = P_e + P_i + P_o$$

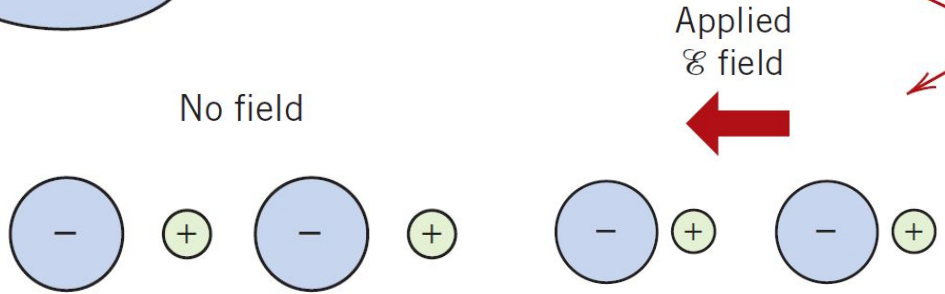
Electronic polarization (P_e)



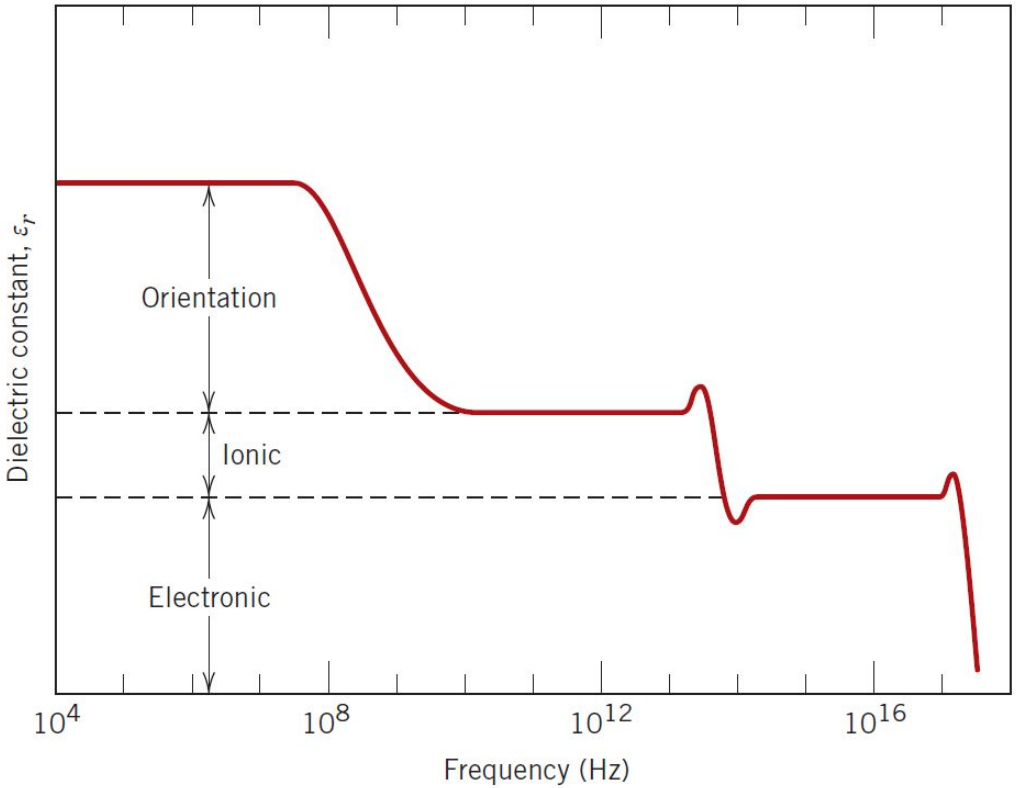
Orientation polarization (P_o)



Ionic polarization (P_i)



Frequency vs ϵ_r



- Dielectric material is subject to polarization by AC electric field
- With each direction reversal, dipoles try to reorient with field
- Process requires finite time → **relaxation frequency** is reciprocal of this minimum reorientation time
- frequency of E > relaxation frequency → dipole cannot react → no contribution to ϵ_r
- When a polarization mechanism ceases to function, there is an abrupt drop in ϵ_r

Dielectric loss and Dielectric Strength

Dielectric loss:

- Absorption of electrical energy by a dielectric material that is subjected to an alternating electric field
- Important at electric field frequencies in the vicinity of the relaxation frequency for each of the operative dipole types for a specific materia
- Low dielectric loss is desired at the frequency of utilization

Dielectric strength:

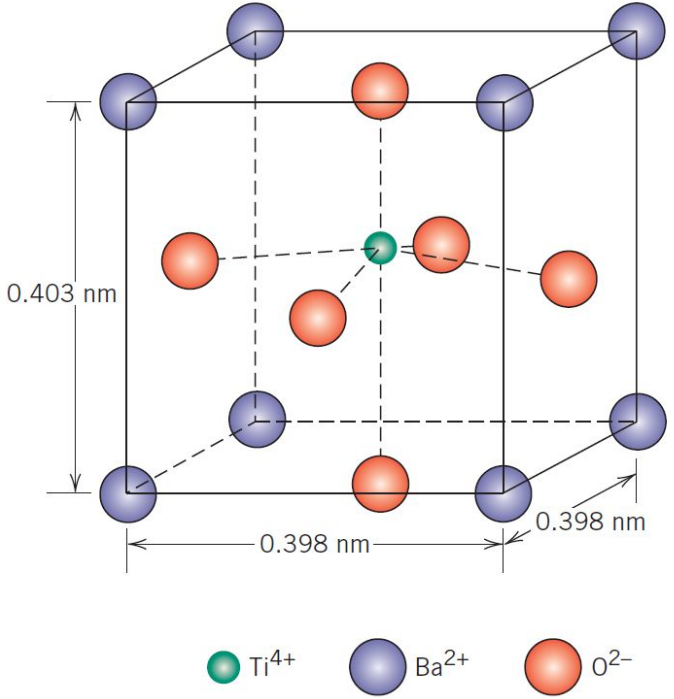
- Magnitude of electric field necessary to produce breakdown (also called breakdown strength)
- High electric field applied across dielectric → large number of electrons excited to conduction band → higher current → localized melting, burning, or vaporization producing irreversible degradation → failure of material
- Higher dielectric strength preferred

Examples of Dielectric

Material	Dielectric Constant		Dielectric Strength (V/mil)[a]
	60 Hz	1 MHz	
Ceramics			
Titanate ceramics	—	15–10,000	50–300
Mica	—	5.4–8.7	1000–2000
Steatite (MgO–SiO_2)	—	5.5–7.5	200–350
Soda–lime glass	6.9	6.9	250
Porcelain	6.0	6.0	40–400
Fused silica	4.0	3.8	250
Polymers			
Phenol-formaldehyde	5.3	4.8	300–400
Nylon 6,6	4.0	3.6	400
Polystyrene	2.6	2.6	500–700
Polyethylene	2.3	2.3	450–500
Polytetrafluoroethylene	2.1	2.1	400–500

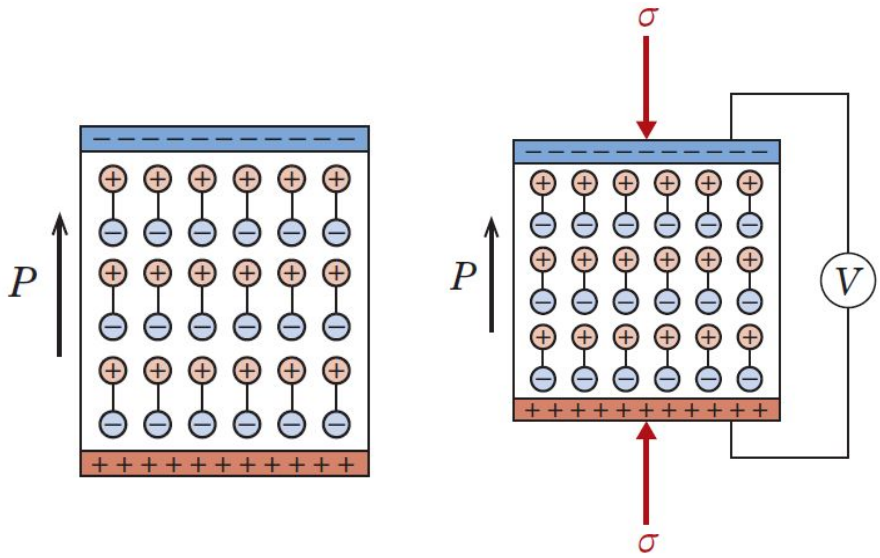
Ferroelectric Materials

- Exhibits spontaneous polarization—polarization in absence of E
- Spontaneous polarization can be reversed by an external electric field
- Crystal structure results in the permanent electric dipoles (**spontaneous polarization**)
- Have extremely high ϵ_r even at low applied field frequencies ($\epsilon_{r\text{-}BaTiO3} = 5000$)
- Application - Sensors and Actuators
- Emerging applications - FeFETs, FRAM



Piezoelectric Materials

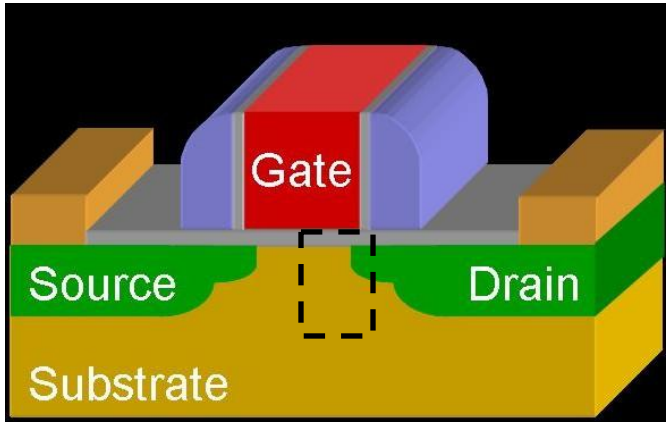
- Electric polarization (i.e., an electric field or voltage) induced as a result of a mechanical strain produced from the application of an external force
- Reversing the sign of the force (e.g., from tension to compression) reverses the direction of the field
- Inverse piezoelectric effect also displayed → mechanical strain due to imposition of electrical field
- Property is characteristic of materials having crystal structures with a lower degree of symmetry than ferroelectrics



- GaN (RF and Power transistors), AlN (RF Filters), lead zirconate–titanate (PZT) (energy harvesters), Quartz (oscillator, timing), sensors, gas lighter..

Metal Oxide Semiconductor (MOS) Structure

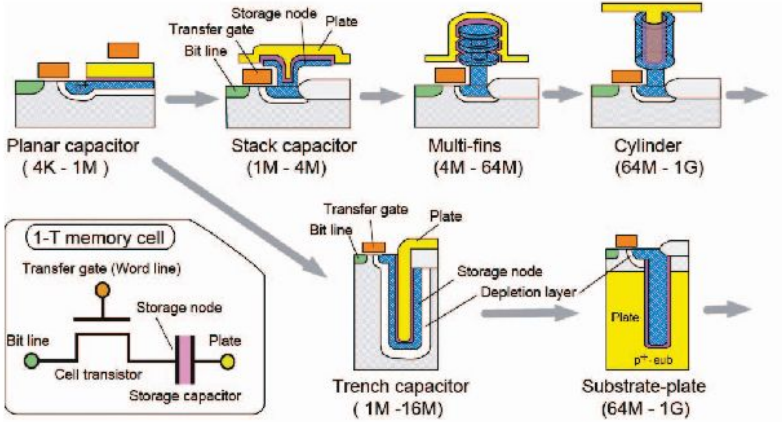
Important part of logic (transistor), memory (DRAM and flash) and Image sensors



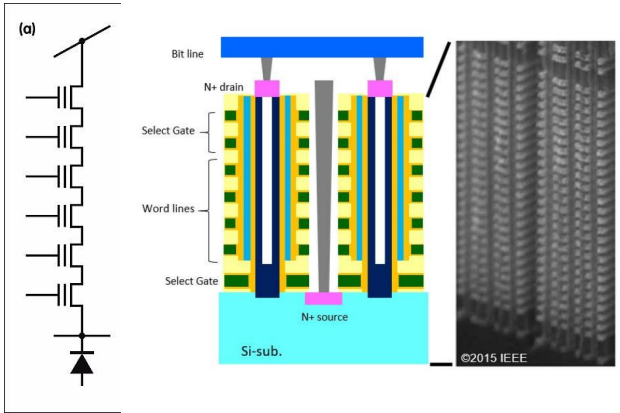
The most abundant man made object on planet earth

Integrity of gate dielectric is of paramount importance

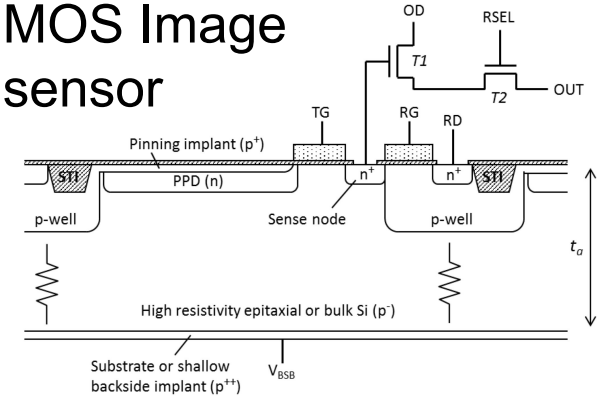
DRAM



NAND Flash

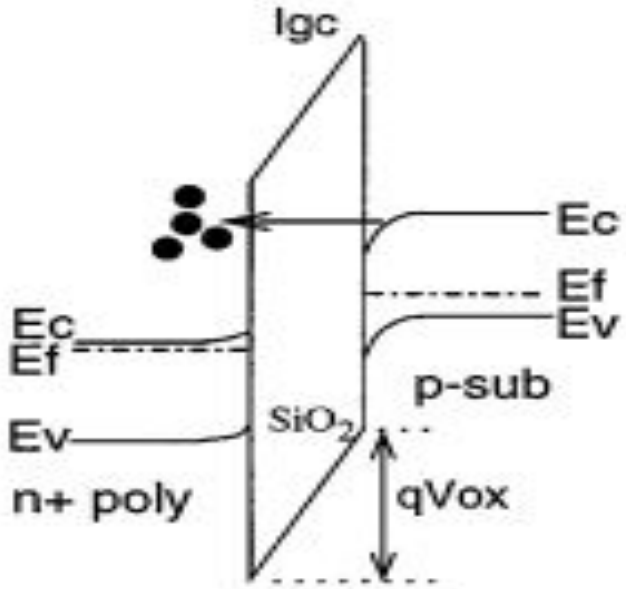


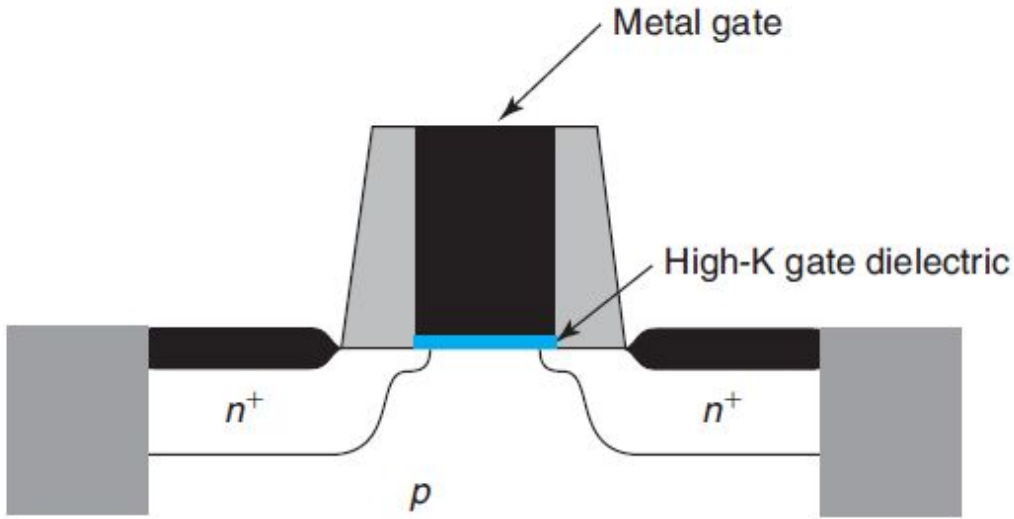
MOS Image sensor



Excessive gate leakage current through ~ 2 nm thin barrier

forced MOS Field Effect Transistor to undergo a heart replacement





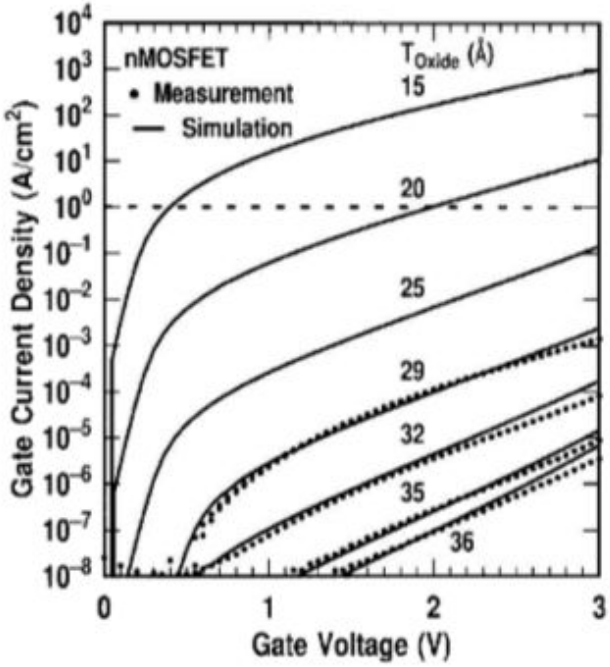
Hf based higher dielectric constant material as gate insulator and metal in place of polysilicon gate (since 45/32 nm in 2007)

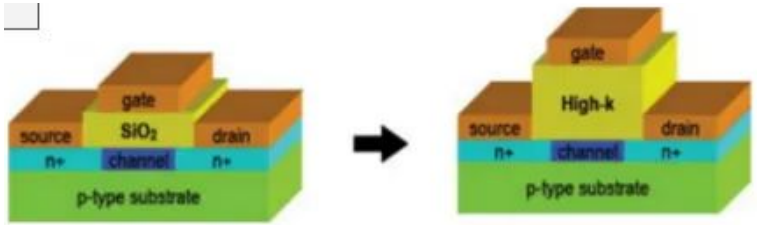
https://spectrum.ieee.org/transistor-history-2670915657

Why High-k gate dielectrics?

$$C = \kappa\epsilon_0 \frac{A}{t}$$

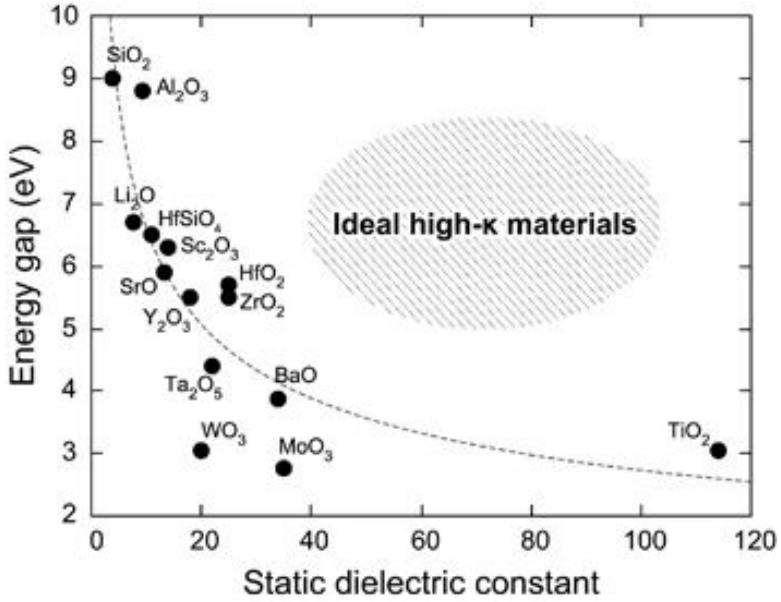
Maintaining C
1) Decrease t_{ox}
2) Use higher k





$$\frac{C}{A} = \frac{\kappa_{SiO_2}\epsilon_0}{t_{SiO_2}} = \frac{\kappa_{hk}\epsilon_0}{t_{hk}}$$

$$EOT = t_{ox} = t_{hk}\frac{\kappa_{SiO_2}}{\kappa_{hk}}$$

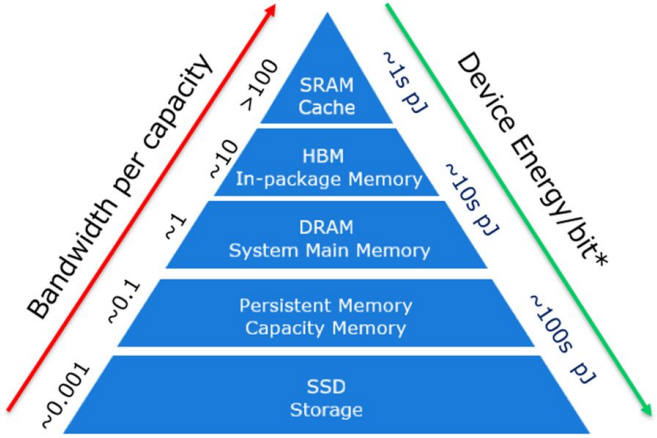


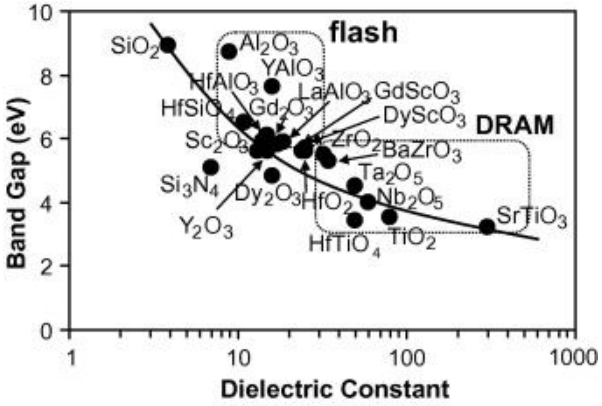
Device scaling

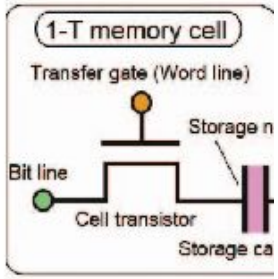
→ t_{oxide} ↓ → I_G ↑

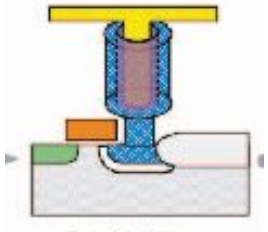
Thicker high-k gate oxide without comprising gate capacitance → better device performance

High-k dielectrics for memory

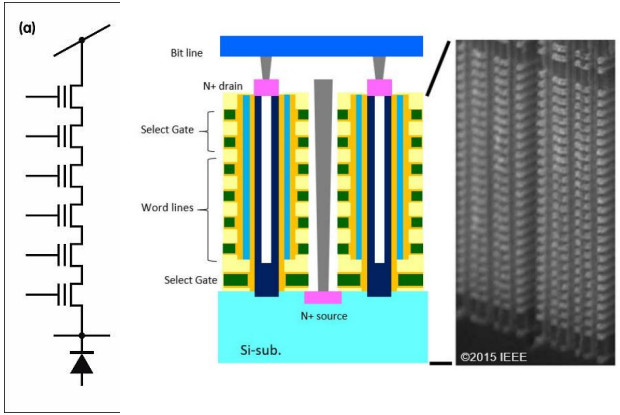








Stacked Dynamic Random Access Memory (DRAM) bit cell



3D NAND Flash array (side and top view)

